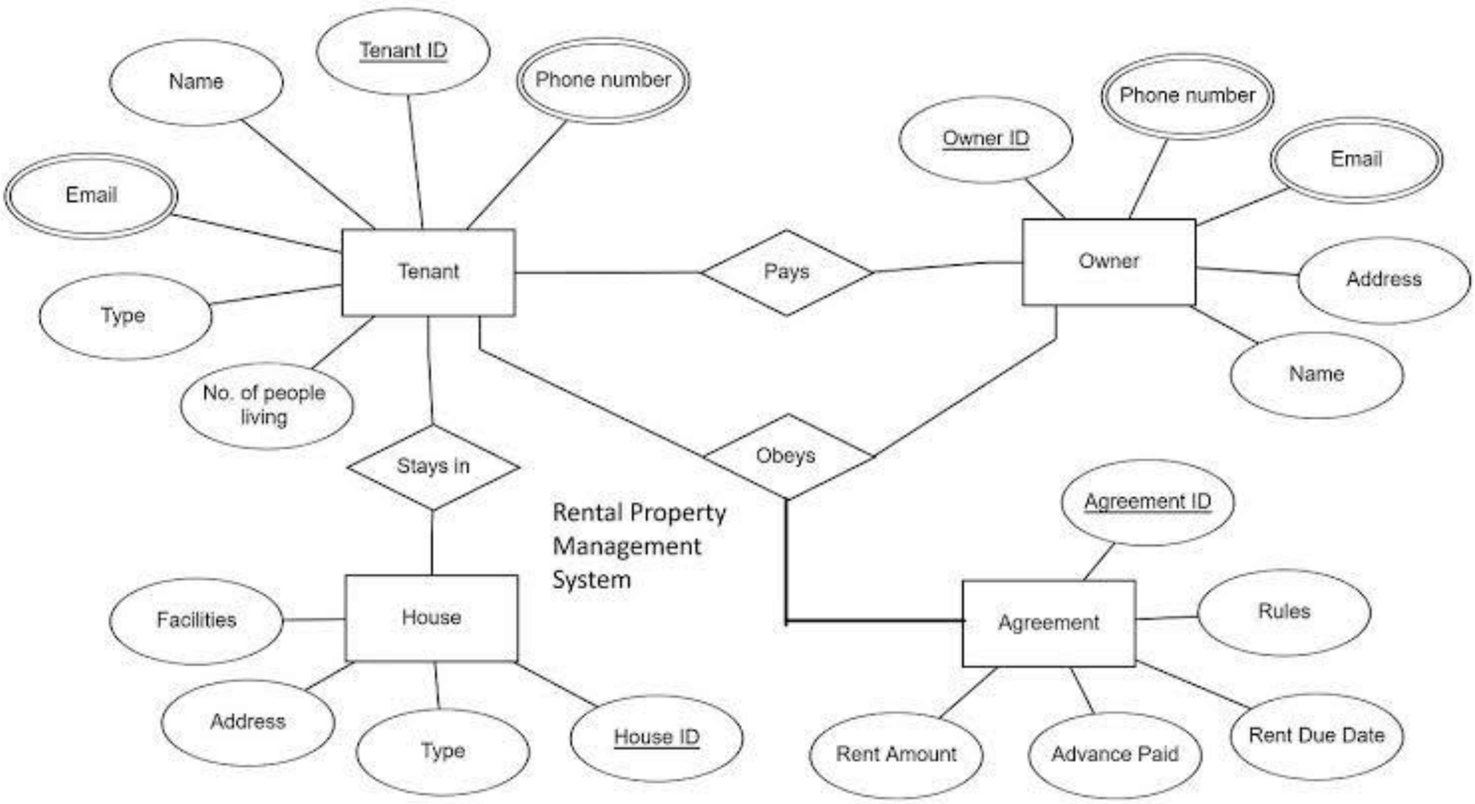
National University of Computer & Emerging Sciences, Karachi
School of Computing Department
Fall 2025, Practice Tasks

Course Code: CL-2005	Course : Database Systems Lab
Instructor(s) :	Fatima Gado

Task-01

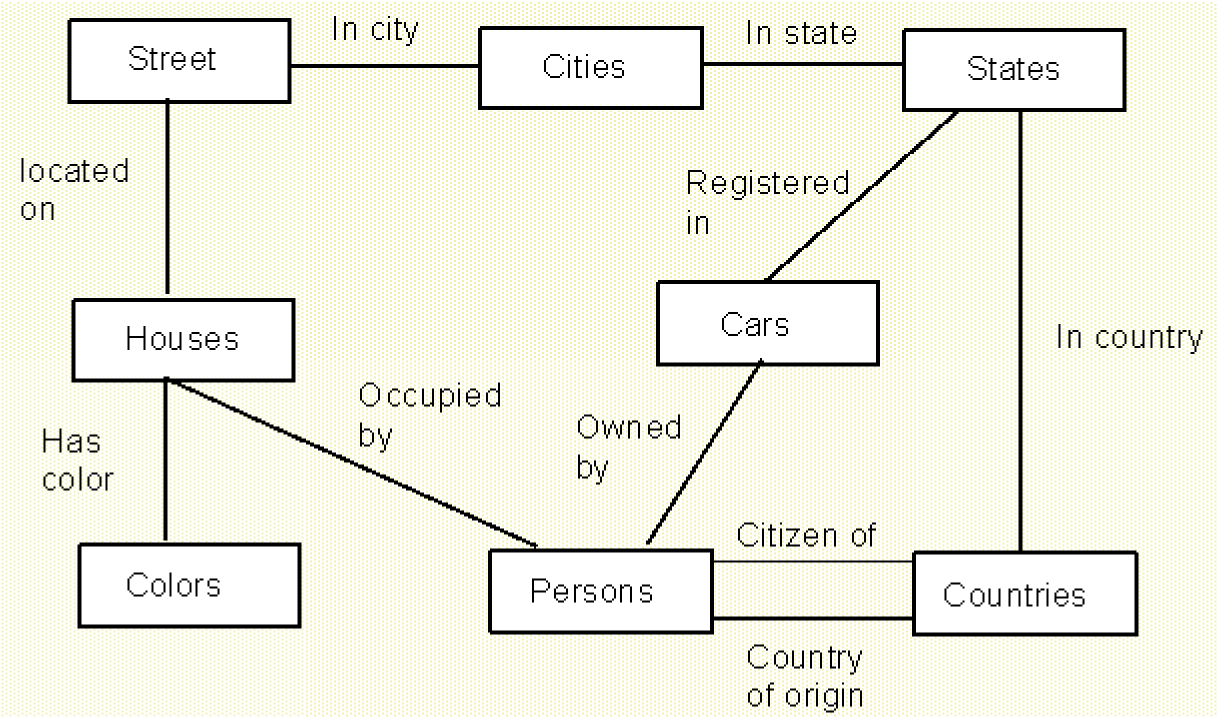


a) Develop the Logical Model Diagram for the CONFERENCE_REVIEWdatabase and build the design using a data modeling tool data modeler

b) Develop the Relational Model Diagram for the CONFERENCE_REVIEW database and build the design using a data modeling tool data modeler.

c) Generate DDL

Task-02

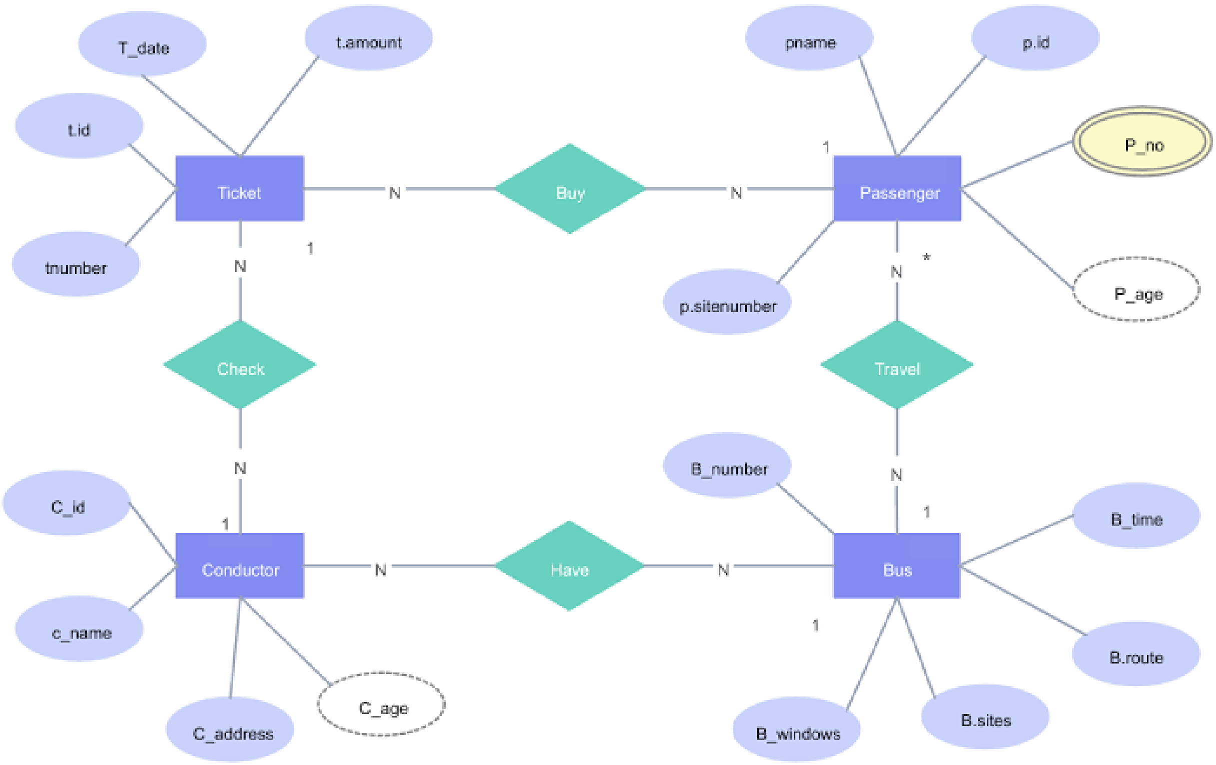


a) **Develop the Logical Model Diagram for the PropertyAndOwnershipDB above database and build the design using a data modeling tool data modeler**

b) **Develop the Relational Model Diagram for the PropertyAndOwnershipDB database and build the design using a data modeling tool data modeler**

c) **Generate DDL**

` Task-03



- Create a physical design (DDL) from the above logically designed database.
- Keeping in mind the logical design, create foreign keys in each table where required.
- Populate each table up to maximum three records.

Task 4

You have been hired by a **National Film Festival Committee** to design a comprehensive database system that manages film submissions, screenings, directors, venues, and judging results. The goal is to create a structured model for efficient data tracking and reporting.

1. **Films:**
Each submitted film has a unique film ID, title, genre, duration, and language. Every film is directed by one or more directors and screened in specific venues during the festival.
2. **Directors:**
Director information includes name, nationality, contact number, and area of expertise (e.g., documentary, short film, feature film). A director can submit multiple films.
3. **Venues:**
The festival uses multiple screening venues, each identified by name, type (indoor, outdoor, auditorium), and capacity. Each venue may host several films during different time slots.
4. **Judges:**
Judges are responsible for evaluating the films. Judge data includes name, experience level, and specialty area.
5. **Evaluations:**
Each judge evaluates several films and records ratings for direction, creativity, and audience impact, along with overall remarks such as “excellent,” “average,” or “not selected.”

Task 5

You have been hired by a **Research Conference Organizing Committee** to design a database system to handle paper submissions, reviewers, sessions, and presentation schedules efficiently.

1.**Researchers:**
Each researcher has a name, email, institution, and field of study. A researcher may submit one or more papers.
2.**Papers:**
Each paper has a title, abstract, topic category, and file name. Every paper is linked to its author and will be scheduled for presentation in a specific session.
3.**Sessions:**
Sessions are identified by session ID, topic, date, and room. Each session includes multiple papers and is managed by a session chair.
4.**Reviewers:**
Reviewers evaluate papers based on criteria such as originality, clarity, and relevance. Reviewer data includes name, email, and area of expertise.
5.**Reviews:**
Each review includes the reviewer's ID, paper ID, rating (1–10), and comments.

Queries

1. Display the employee names and department names where the employee earns the maximum salary in their respective department.
2. Find all employees whose job title matches the job title of at least one employee in the Finance department.
3. List the departments where no employees have a commission percentage assigned.
4. Display employees who were hired in the same year as their manager. Include employee name, manager name, and hire year.
5. Show the department name and total number of employees for departments located in countries starting with the letter 'U'.
6. Display the job titles and total number of employees where the average salary is greater than 10,000 and at least one employee earns below 8,000.
7. Retrieve the list of employees whose salary is greater than the average salary of all employees in the same region.
8. Display employee names who work in departments that share the same location as the Sales department.
9. List the region name, country name, and total number of departments in each country that has more than one department.
10. Display employees who don't have any subordinates (i.e., no one reports to them), along with their department and job title.